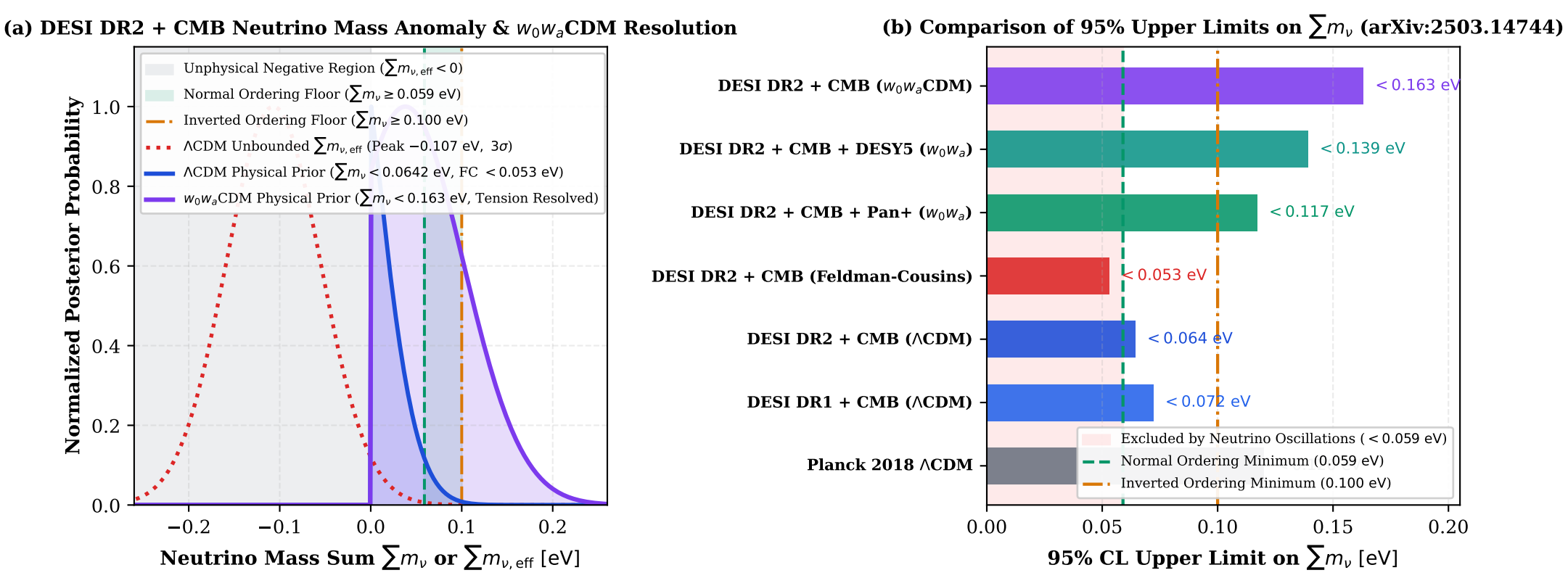

(a) DESI DR2 + CMB Neutrino Mass Anomaly & w_0w_aCDM Resolution
Unphysical Negative Region ($\sum m_{\nu,\mathrm{eff}} < 0$)
Normal Ordering Floor ($\sum m_\nu \geq 0.059$ eV)
Inverted Ordering Floor ($\sum m_\nu \geq 0.100$ eV)
ΛCDM Unbounded $\sum m_{\nu,\mathrm{eff}}$ (Peak −0.107 eV, 3σ)
ΛCDM Physical Prior ($\sum m_\nu < 0.0642$ eV, FC < 0.053 eV)
w_0w_aCDM Physical Prior ($\sum m_\nu < 0.163$ eV, Tension Resolved)
Normalized Posterior Probability
0.0
0.2
0.4
0.6
0.8
1.0
−0.2
−0.1
0.0
0.1
0.2
Neutrino Mass Sum $\sum m_\nu$ or $\sum m_{\nu,\mathrm{eff}}$ [eV]
(b) Comparison of 95% Upper Limits on $\sum m_\nu$ (arXiv:2503.14744)
DESI DR2 + CMB (w_0w_aCDM)
< 0.163 eV
DESI DR2 + CMB + DESY5 (w_0w_a)
< 0.139 eV
DESI DR2 + CMB + Pan+ (w_0w_a)
< 0.117 eV
DESI DR2 + CMB (Feldman-Cousins)
< 0.053 eV
DESI DR2 + CMB (ΛCDM)
< 0.064 eV
DESI DR1 + CMB (ΛCDM)
< 0.072 eV
Planck 2018 ΛCDM
Excluded by Neutrino Oscillations (< 0.059 eV)
Normal Ordering Minimum (0.059 eV)
Inverted Ordering Minimum (0.100 eV)
0.00
0.05
0.10
0.15
0.20
95% CL Upper Limit on $\sum m_\nu$ [eV]